

Name: _____

Class: _____

ACTIVITY SHEET

Chapter 6: Prior knowledge

Decide whether each of the following statements is true or false. If the statement is false, correct it.

- 1 The nucleus is the centre of positive charge due to the presence of protons. (True / False)

- 2 Electrons are high-speed particles held in specific shells around the nucleus due to the electrostatic force of attraction that the nucleus exerts upon them. (True / False)

- 3 Electrons can only occupy specific shells and sub-shells around the nucleus if they have the right amount of energy. (True / False)

- 4 Electrons in a particular subshell all have different amounts of energy. (True / False)

- 5 A sub-shell has a specific number of orbitals, each of which can be occupied by either a single electron or two electrons of opposite spin. (True / False)

- 6 Valence electrons are those found in the inner occupied shell of an atom. (True / False)

- 7 Atoms are energetically stable when they have a full valence shell and, because this often requires eight electrons, it is known as the octet rule. (True / False)

- 8** Electrons in a particular subshell fill orbitals individually until they are forced to share an orbital, and only then if they have the same spin. (True / False)
- 9** Electron dot formulas of atoms and molecules show only valence electrons and whether they are found singly, paired up or shared by two atoms. (True / False)
- 10** Valence structure diagrams use a line to show a pair of electrons shared by two atoms (bonding pair) and dots to show non-bonding electrons. (True / False)
- 11** Covalent bonding occurs when a pair of electrons is shared between two atoms because each gains a less energetically stable state. (True / False)
- 12** The force that holds the atoms in a molecule together is the electrostatic force of attraction that the positively charged nuclei have for the shared negatively charged electrons. (True / False)
- 13** Each pair of shared electrons is called a bonding pair, and each pair of unshared electrons is called a non-bonding or lone pair. (True / False)
- 14** A single bond occurs when a pair of atoms shares one electron, a double bond occurs when they share two, and a triple bond occurs when they share three. (True / False)
- 15** Ball-and-stick molecular models show the bonds between atoms, while space-filling models show the volume and placement of atoms relative to each other. (True / False)

- 16** Intramolecular forces are the electrostatic forces of attraction that hold the atoms within a molecule together. (True / False)
- 17** Intermolecular forces are the forces of repulsion that occur between the individual particles (molecules or ions) of a substance. (True / False)
- 18** Electronegativity refers to the ability of an ion to attract electrons toward itself in a chemical bond. (True / False)
- 19** A solvent is the substance that dissolves in a solute to form a solution. (True / False)
- 20** An aqueous solution is a solution in which the solvent is water. (True / False)

Having difficulty?

Revisit earlier chapters of your textbook, particularly sections 1.6 and 1.7 of *Chapter 1, Atoms and elements*, and section 3.5 of *Chapter 3, Metallic, ionic and covalent structure bonding*.